



Value in Vogue (*VinV*) – Applied Value Behavior Under Macro Regimes

*Downstream Proof of Concept
(Non-Forecasting, Descriptive)*



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Purpose & Boundary

VinV is a **downstream applied research layer, explicitly separated** from **FT-GMI & free of forecasting or allocation claims.**

02

Existing Data Artifacts

Summarizes the **canonical equity-macro panel, returns integration, & evaluation machinery** already built.

03

Regime Conditioning

All analysis is **segmented by macro regimes** rather than pooled across **full samples.**

04

Behavior Under Stress

Descriptively compares VinV with **related benchmarks** across **drawdown & recovery windows.**

05

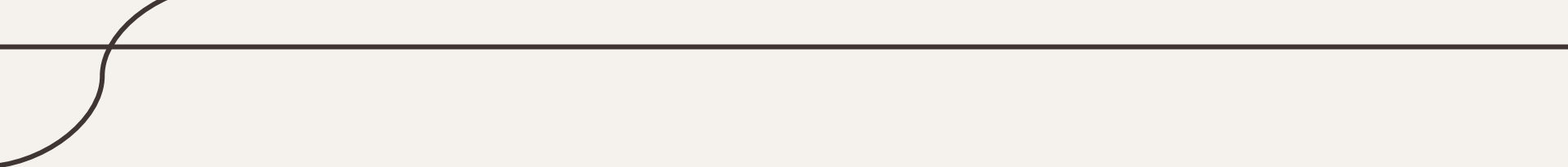
Observed Structure

Highlights **dispersion, volatility, persistence, & recovery slope** as **behavioral properties, not signals.**

06

Completion Path

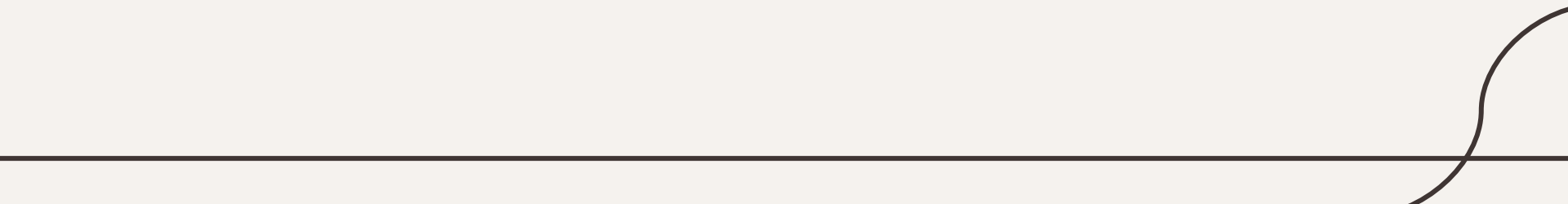
States what **VinV is not yet & how it completes** after **FT-GMI is finalized** without **contaminating diagnostics.**



01

Purpose & Boundary

Defines VinV's scope, intent, & explicit non-claims to enforce clear analytical boundaries from the outset.



System Context & Governance Boundary –

Layered macro-financial research architecture separating data, diagnostics, interpretation, & application.



the Spine (Foundational Layer)

- **Canonical macro-financial data fusion & normalization,**
- **G20-scoped macro-financial inputs** across rates, FX, inflation, credit, commodities, equities, & real economy metrics,
- **Deterministic feature contracts with locked semantics,**
- **Canonical raw data stored** in governed **cloud object storage (Cloudflare R2);** **read/write access restricted to the Spine layer only,** &
- Infrastructure for **reproducibility, auditability, & scale.**



Downstream Diagnostic & Application Layers

- **FT-GMI** — regime-aware macro-financial diagnostics (*read-only*),
- **VinV** — U.S. equity behavior conditioned on macro regimes, &
- **the OracleChambers** — interpretive narrative layer (*non-authoritative*).

AI/ML tooling constrained to **monitoring, explanation, & validation support** — operating strictly downstream ("*living off the exhaust*") **within the OracleChambers.**

FT-GMI Introduction –

Regime-aware macro-financial diagnostics layer

What is FT-GMI?

- It measures **structural macro-financial stress within the G20**, not outcomes,
- Its **outputs define regimes & stress windows** used by VinV,
- FT-GMI is **read-only & never altered by downstream analysis**,
- **FT-GMI consumes canonical macro-financial data produced by the Spine (fusion & normalization layer)**, &
- **the Spine** is the **canonical fusion, normalization, & governance layer** supplying all macro-financial inputs to FT-GMI.



Relationship FT-GMI & VinV

- **VinV** is a **U.S.-focused downstream layer** within the **FT-GMI layer**,
- **VinV conditions equity behavior** on the **macro context** provided by **FT-GMI**, &
- The dependency is strictly one-way ($FT-GMI \rightarrow VinV$).

VinV: Why This Exists

Purpose Statement

VinV is a **downstream applied research layer** designed to *describe how value-oriented equity constructions behave under different macro regimes.*

What VinV Does

- Examines **historical behavior** of value baskets,
- Operates **conditionally on macro regimes**, &
- Uses **descriptive comparisons**, not predictions.

What VinV Is For

- Understanding **durability, dispersion**, & **recovery** characteristics,
- Translating **macro diagnostics** into **observed asset behavior**, &
- Demonstrating **applied execution capacity**.

VinV does not forecast, time, optimize, or recommend allocation.

Boundary

Boundary Rules

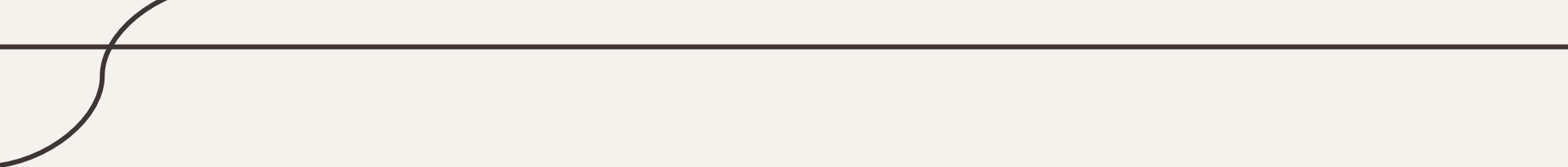
- VinV consumes upstream outputs **read-only**,
- VinV **cannot** tune, validate, or alter FT-GMI, &
- VinV success or failure **does not affect** diagnostics.

Supervisory Implication

- **No forecasting oversight** requested,
- **No performance endorsement** implied, &
- **No** expansion of supervisory scope.



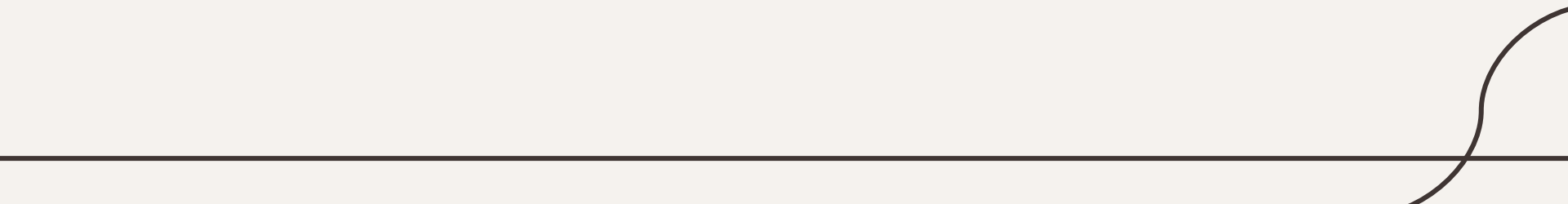
VinV exists to show applied execution, not to extend diagnostic authority.



02

Existing Data Artifacts

Documents the governed data artifacts already constructed, independent of modeling or forecasting assumptions.



Existing Data Artifacts (Overview)

What Exists Today (Concrete, Not Aspirational)

Artifacts in place

- **Canonical equity-macro panel,**
- **Returns integration & validation pipeline,**
- **Regime-aware temporal structure, &**
- **Defensive data contracts & checks.**

These artifacts exist independently of any model choice or forecasting logic.

Canonical Equity–Macro Panel

Long-Horizon, Regime-Compatible Panel

Panel characteristics

- **Monthly frequency, symbol–date indexed,**
- Clean **equity identifiers & returns,**
- **Time-aligned macro variables, &**
- **Stable schema & versioned inputs.**

Design intent

- Enable **cross-sectional comparisons,**
- Support **regime segmentation, &**
- Avoid ad-hoc feature construction.

No raw provider data enters VinV directly.

Returns Integration & Validation

Returns Are Explicitly Governed

What is implemented

- **Normalized monthly returns** (*ret_m*),
- **Column presence & type validation**,
- **Defensive joins** (*no silent drops*), &
- **Symbol/date sorting & integrity checks.**

Why this matters

- **Eliminates common backtest leakage**,
- Makes **downstream evaluation auditable**, &
- **Separates data hygiene** from **modeling**.

Feature Construction Layer

Descriptive, Cross-Sectional Features

Feature families

- **Rank-based measures,**
- **Z-score normalization,**
- **Dispersion & breadth metrics, &**
- **Persistence & churn indicators.**

Features describe relative structure, not expected payoff.

Regime-Aware Evaluation Windows

Time Is Treated as Conditional

What exists

- **Regime-segmented analysis windows** defined upstream by FT-GMI,
- **Explicit stress-period slices**, &
- **Post-stress recovery windows**.

Why

- **Avoids full-sample averaging**,
- Treats **crises as information, not noise**, &
- Aligns with **applied macro time-series logic**.

Artifact Independence (Boundary Reinforcement)

Artifacts Are Upstream-Stable

Key property

- All **VinV artifacts remain valid** if:
 - **FT-GMI outputs** are **removed**,
 - **Models** are **changed**, &
 - **Forecasting** is **never added**.

The data foundation does not assume, require, or imply prediction.



03

Regime Conditioning

Positions macro regimes as contextual states for interpretation rather than objects of prediction or timing.



Regime Conditioning (Why It Exists)

Why Regime Conditioning Is Non-Optional

Core premise

Macro-financial behavior is **state-dependent**, not stable across time.

Design stance

- **Full-sample averages** are **misleading**,
- **Structural breaks** are **information**, not noise, &
- **Stress periods** are **analytically distinct**.

VinV treats regime shifts as first-class conditions, not exceptions.

What a “Regime” Means Here

Regimes Are Contextual, Not Predictive

Regime definition *(operational)*

- A regime is a **persistent macro environment** characterized by:
 - **Constraint tightness,**
 - **Liquidity conditions, &**
 - **Stress transmission intensity.**

Clarification

- Regimes are **not forecasts,**
- Regimes are **not labels for timing, &**
- Regimes are used solely to **condition observation.**

Regime identification does not imply regime transition prediction.

Temporal Segmentation Logic

How Time Is Treated

Segmentation approach

- **Long-run baseline context,**
- **Financialization era context, &**
- **Stress-intensive sub-windows.**

Why this matters

- **Prevents dilution of crisis behavior,**
- **Avoids smoothing away nonlinear responses,**
&
- **Preserves interpretability under stress.**

Time is segmented to observe behavior, not to optimize outcomes.

Conditional Evaluation (Not Pooling)

What Conditioning Enables

With regime conditioning

- **Like-for-like comparisons,**
- **Cross-sectional behavior** under **shared constraints**, &
- **Meaningful stress** vs **non-stress contrasts.**

Without conditioning

- **False stability,**
- **Misleading averages,** &
- **Overconfident conclusions.**

VinV does not pool incompatible macro environments.

Relationship to Diagnostics (Boundary Reinforced)

How This Relates to FT-GMI

Clarification

- **FT-GMI diagnoses macro structure,**
- **VinV conditions on macro context,**
- **No diagnostic output is altered by VinV, &**
- **Diagnostic immutability is enforced by design,** not convention.

Conditioning uses macro context as a lens, not as a target.

Why This Matters for Interpretation

Interpretive Discipline

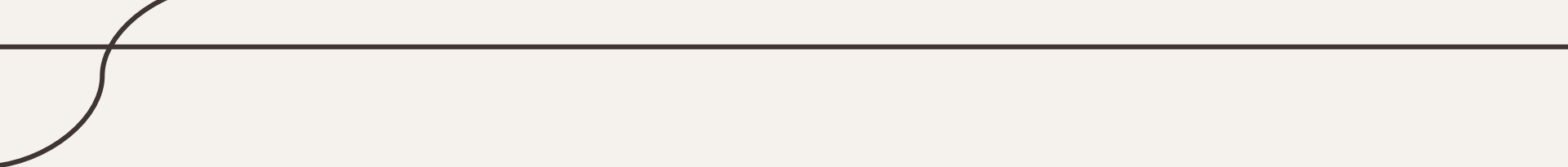
Regime conditioning allows

- **Honest statements** such as:
 - “**Behavior differs under stress**”, &
 - “**Recovery paths are regime-specific.**”

What it prevents

- **Universal claims,**
- **Overgeneralization, &**
- **Implied prediction.**

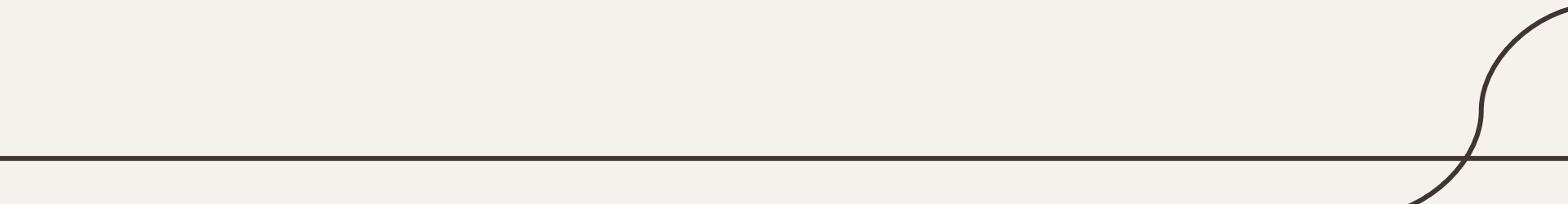
Regime conditioning narrows claims to what the data can support.



04

Behavior Under Stress

Examines how equity structures behave under macro stress without asserting performance, signals, or tradability.



Global Comment

(Applies to ALL Graphs in This Section)

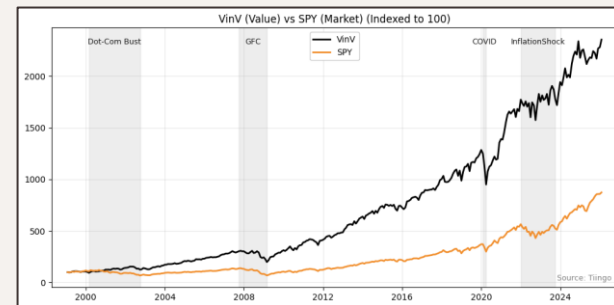
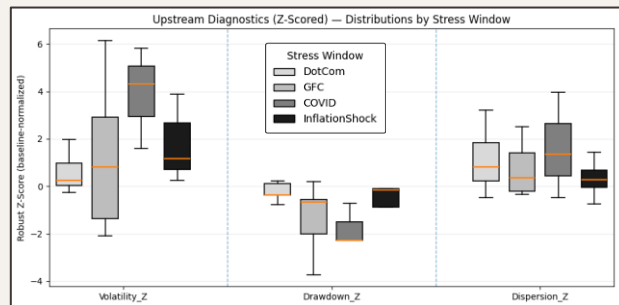
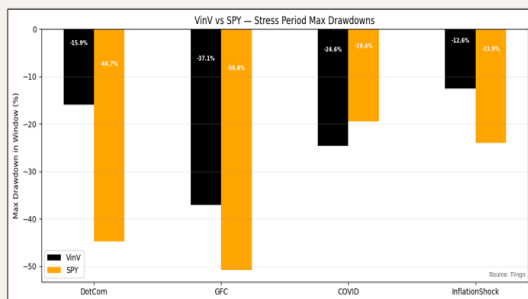
All figures are **descriptive, regime-conditional comparisons** of VinV relative to standard benchmarks during identified macro stress windows.

Graphs are **indexed to a common starting point, conditioned on shared stress regimes**, & intended to illustrate **structural behavior (drawdown, volatility, dispersion, recovery shape)**, **not prediction, timing, allocation, or performance claims**.

All stress windows shown correspond to **historically recognized U.S. macro dislocations** (*Dot-Com, GFC, COVID, Inflation Shock*).

Global Comment – Stress Periods

Stress periods are inputs, not conclusions.



Stress Period	Dates	Why This Period Qualifies as Stress (Minimal, Factual)	What FT-GMI Uses It For
DotCom Bust	Mar 2000 – Oct 2002	Equity valuation collapse following capital misallocation & tightening financial conditions	Regime contrast, equity fragility, recovery asymmetry
Global Financial Crisis (GFC)	Jul 2007 – Mar 2009	Systemic credit contraction, liquidity seizure, synchronized global drawdown	Liquidity transmission, policy constraint diagnostics
COVID Shock	Feb 2020 – Apr 2020	Sudden exogenous halt in economic activity with extreme volatility & liquidity stress	Shock absorption vs amplification behavior
InflationShock	Jan 2022 – Oct 2023	Inflation regime shift with aggressive global monetary tightening & valuation compression	Inflation constraint regimes, cross-asset stress interaction

Stress period definitions are fixed upstream by FT-GMI and consumed read-only by VinV.

VinV vs SPY –

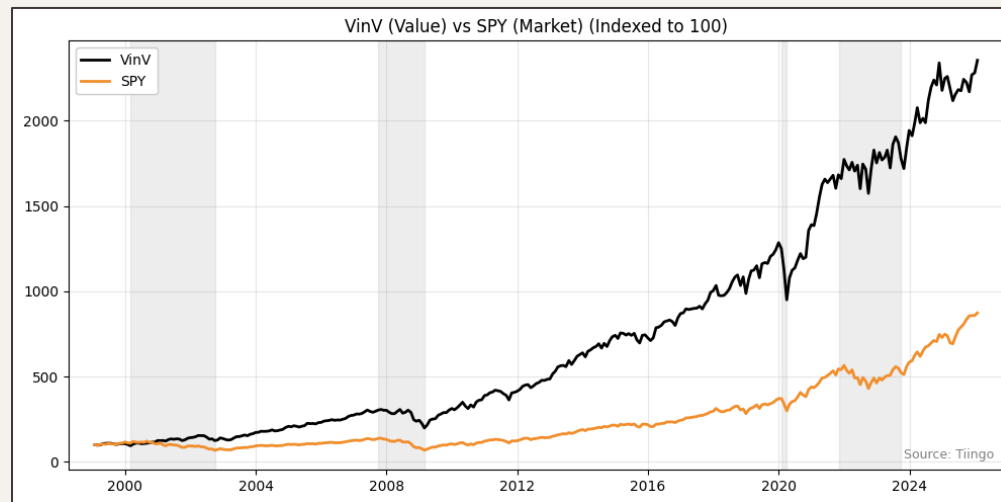
Long-Horizon Capital Trajectories (Indexed)

Comment

- **VinV** reflects the aggregated capital path of **dividend-persistent equities** constructed under **fixed inclusion rules**, &
- **SPY** serves as a **broad-market reference** for **regime-level comparison**, not **performance benchmarking**.

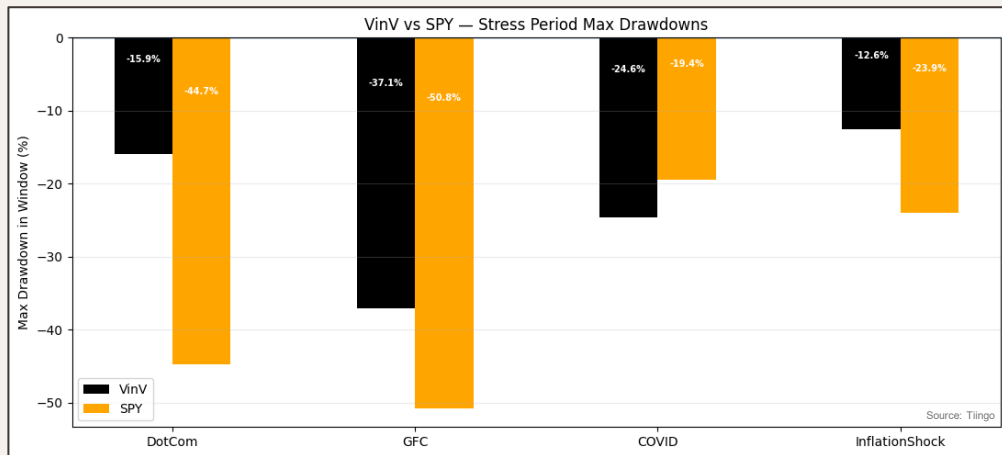
What it is NOT

- **Not** an **outperformance** or **alpha comparison**,
- **Not** a **return forecast** or **expectation**, &
- **Not** a **trading** or **allocation signal**.



VinV vs SPY —

Maximum Drawdowns Across Major Stress Regimes



Comment

- **VinV** exhibits **materially different drawdown depth** across **stress regimes** relative to the **broad market**, reflecting **structural composition** rather than **defensive positioning**, &
- The comparison **isolates loss severity** within **defined stress windows**, **independent of recovery timing** or **post-crisis performance**.

What it is NOT

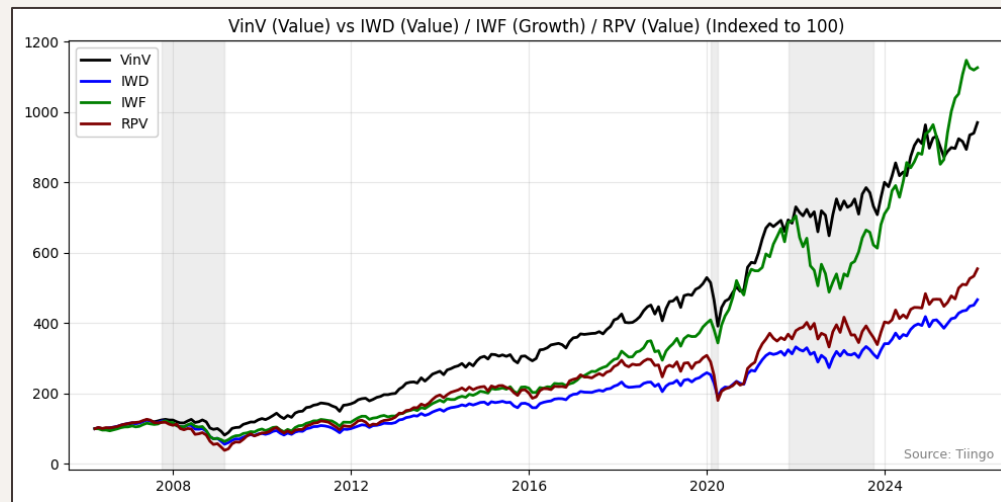
- **Not** a **recovery timing** tool,
- **Not** a **forecast of normalization speed** or **policy response**, &
- **Not** a **fitted model** or **optimized portfolio**.

VinV vs Benchmarks –

Long-Horizon Capital Paths Across Value & Growth Constructions (Indexed)

Comment

- **VinV** reflects the **long-horizon capital path** of a **fixed, value-oriented equity construction** evaluated under **identical market history** as **standard style benchmarks**, &
- **IWD, IWF, & RPV** serve as **reference constructions** representing **conventional value & growth exposures**, included to **contextualize structural differences in compounding behavior** rather than to benchmark performance.

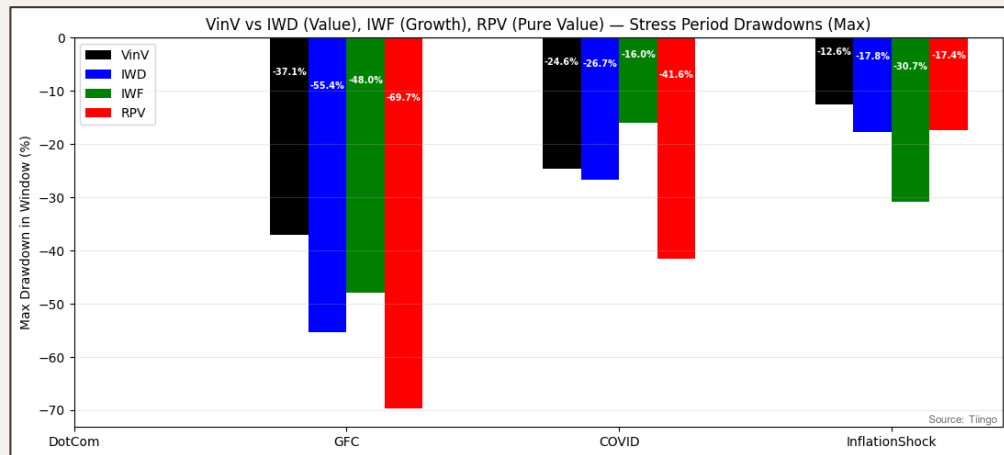


What it is NOT

- **Not** an **outperformance, alpha, or factor dominance** claim.
- **Not** a **return forecast, expected payoff, or capital allocation recommendation**, &
- **Not** evidence of **robustness outside the defined constructions or historical window shown**.

VinV vs Benchmarks –

Relative Drawdown Comparison (Showing Depth & Period)



Comment

- Across stress regimes, VinV exhibits a **distinct drawdown profile** relative to **style benchmarks**, reflecting **portfolio construction** & **factor exposure** rather than defensive positioning or active risk mgmt., &
- Observed differences are **regime-specific** & do not imply consistency of behavior outside defined stress windows.

Why this is allowed

- Drawdown is treated as a risk characteristic, not a payoff.

Stress Window Diagnostic Scorecard –

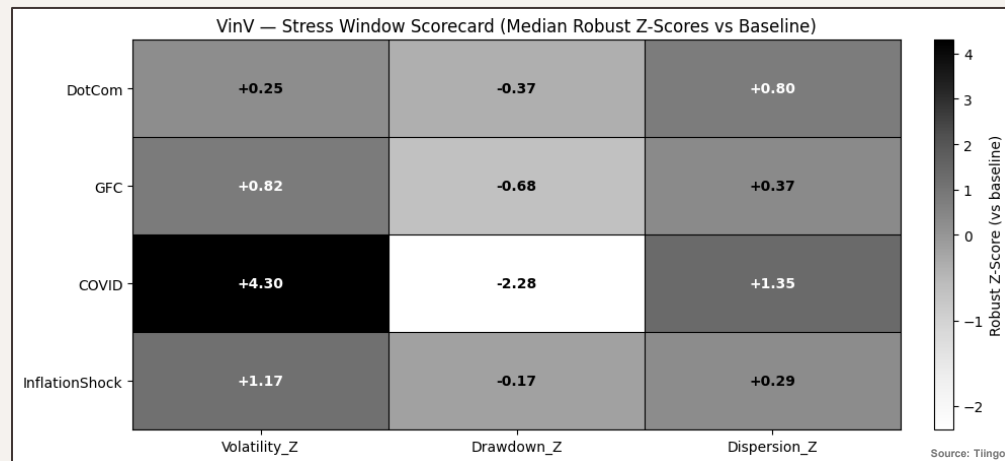
Median Robust Z-Scores vs Long-Run Baseline (Regime-Conditioned)

Comment

- **Summarizes VinV's stress behavior across regimes** using volatility, drawdown, & dispersion deviations from baseline.
- **Uses robust Z-scores (median / IQR)** to remain valid under non-Gaussian stress conditions.
- **Enables regime-specific comparison** without averaging across episodes.

What it is NOT

- **Not a performance or alpha ranking.**
- **Not a forecasting, timing, or allocation signal.**
- **Not a causal claim** about macro drivers.



Diagnostics shown as robust Z-scores (median / IQR), normalized to long-run baseline; **volatility** reflects rolling dispersion, **drawdown** reflects rolling peak-to-trough loss, & **dispersion** reflects cross-sectional spread.

VinV – Methodological Guardrails

(Non-Negotiable)

Comment

- **VinV** results shown here are **descriptive** & **conditional** on the **specified windows** & **definitions**.
- **Outputs** are **not** used to generate **forecasts**, **timing rules**, or **allocation directives** in this deck.

What it is NOT

- Not an **allocation engine**
- Not an **optimization workflow**, &
- Not **causal attribution** (no “because” language).

All comparisons are observational; downstream extensions are explicitly deferred.

Behavioral Summary Slide

Permitted summary language

- **Stress impacts** are **regime-dependent**,
- **Drawdown & recovery paths differ structurally**, &
- **Volatility & dispersion respond asymmetrically**.

Explicit non-claims

- **No dominance**,
- **No stable ranking**, &
- **No predictive inference**.

VinV – Deferred Completion Path

(Post FT-GMI Freeze)

Comment

- VinV is intentionally presented here as a **bounded proof-of-concept supporting FT-GMI credibility** via **execution discipline**.
- After **FT-GMI** is **frozen**, VinV will expand into a separate, explicitly **forecast-oriented track** with dedicated **validation & governance**.

What it is NOT

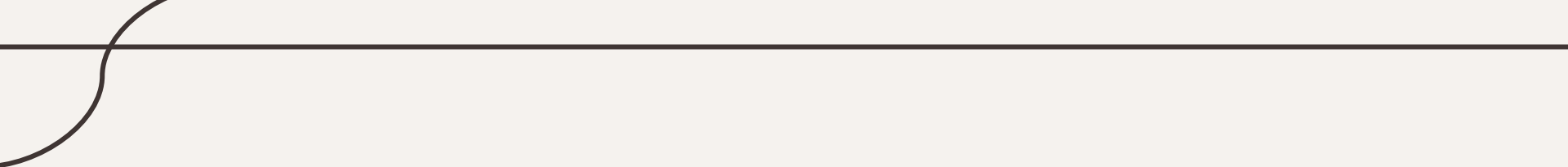
- **Not** a claim that these steps are **completed**,
- **Not** a **performance-marketed** product, &
- **Not** an **implied trading system**.

Planned

(Not Yet Implemented)

- **Benchmark suite freeze & reproducibility manifest**,
- **Walk-forward evaluation** design (**out-of-sample**),
- **Factor attribution & sensitivity analysis**, &
- Reporting layer appropriate for **quant research roles**.

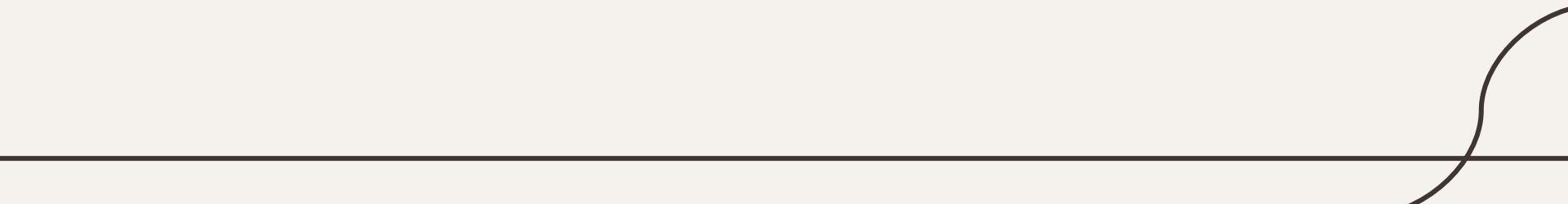
All comparisons are observational; downstream extensions are explicitly deferred.



05

Observed Structure

*Characterizes post-stress recovery shape & persistence as descriptive path properties,
not indicators of timing, signals, or allocation.*



Observed Structure (Bounded Summary)

Observed Structure: Descriptive Patterns Only

Purpose

Summarize **recurring structural characteristics** observed across stress regimes **without inference, ranking, or prediction**.

Observed properties

- **Regime dependence** — behavior varies materially across macro environments,
- **Asymmetric drawdowns** — depth & duration differ across constructions,
- **Nonlinear recovery paths** — recoveries are uneven & path-dependent, &
- **Volatility clustering** — instability concentrates during stress regimes.

These observations describe how structures behaved, not why or what comes next.

Interpretive Limits (Explicit Non-Claims)

What Is Intentionally Not Inferred

From the Observed Structure

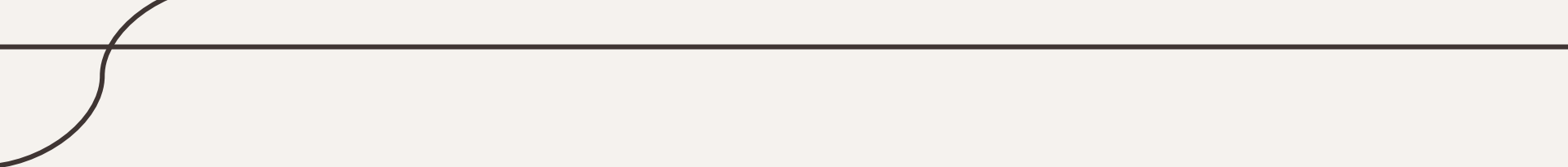
VinV **does not infer**:

- **Causality,**
- **Predictive signals,**
- **Stable rankings or dominance,**
- **Allocation or timing implications, &**
- **Regime transition forecasts.**

Reason

These inferences exceed what descriptive, regime-conditional evidence can support.

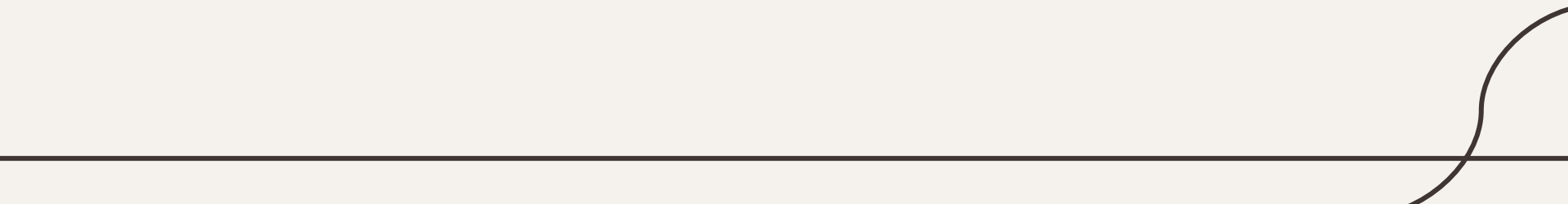
Observed structure represents the ceiling of responsible interpretation at this stage.



06

Completion Path

Synthesizes observed stress & recovery behaviors into a bounded empirical profile, explicitly defining interpretive limits & excluding forecasting, signaling, or allocation claims.



Completion Path (Section Framing)

Completion Path: Sequencing, Not Expansion

Core premise

- **VinV** is **intentionally incomplete** while **FT-GMI** is being finalized, &
- This is a **sequencing decision, not a capability gap**.

Completion occurs by unlocking downstream work, not by revisiting upstream diagnostics.

What “Completion” Means for VinV

Definition of VinV Complete (Applied, Not Marketed)

Completion

VinV is considered **complete** when it includes:

- **Fixed feature contracts** from **FT-GMI** & **the Spine**,
- **Finalized regime segmentation**,
- **Fully specified evaluation windows**,
- **Documented forecasting & benchmarking logic**, &
- **Reproducible results artifacts**.

Completion does not change FT-GMI or its diagnostics.

Why FT-GMI Must Be Fixed First (Diagnostic Immutability)

Upstream Stability Enables Downstream Rigor

FT-GMI finalization provides

- **Stable macro diagnostic definitions,**
- **Locked regime context, &**
- **Non-moving feature semantics.**

Why this matters

- **Prevents re-training on shifting definitions,**
- **Eliminates silent leakage, &**
- **Preserves interpretability.**

Diagnostic immutability is a prerequisite for applied evaluation.

What Unlocks Once FT-GMI Is Final

Downstream Additions (Explicitly Deferred)

Downstream Additions (Explicitly Deferred)

Once FT-GMI is finalized, VinV adds:

- **Forecasting models** (*supervised; statistical & ML methods explicitly scoped*),
- **Benchmark comparisons** (explicit baselines),
- **Cross-regime validation,**
- **Stress-aware performance evaluation, &**
- **Applied research reporting artifacts.**

These additions occur entirely downstream & do not alter diagnostics.

Why This Sequencing Is Deliberate

Risk Management by Design

This sequencing prevents

- **Back-propagation** of outcomes into diagnostics,
- **Overfitting** to regime definitions, &
- **Premature performance claims.**

This sequencing enables

- **Cleaner empirical interpretation,**
- **Clear academic boundaries, &**
- **Stronger applied credibility** later.

Supervisor Impact (Explicitly Limited)

Why This Does Not Expand Supervisory Scope

No supervision requested for

- **Forecasting choices,**
- **Model selection,**
- **Performance evaluation, &**
- **Optimization or trading logic.**

Supervisory role remains

- **Macro framing,**
- **Regime logic, &**
- **Stress interpretation sanity checks.**

VinV completion does not add supervisory burden.

End-State Snapshot

End State (Clean Separation Preserved)

Properties of the end state

- **Diagnostics** remain **non-forecasting**,
- **Application layer** becomes **fully specified**, &
- **No feedback loops introduced.**



Why This Path Matters

By fixing diagnostics first & completing application second, VinV becomes empirically strong without compromising academic integrity.

Thank You

Rand Sobczak Jr.

M.S. Data Science | University of Pittsburgh, **2026 exp.**

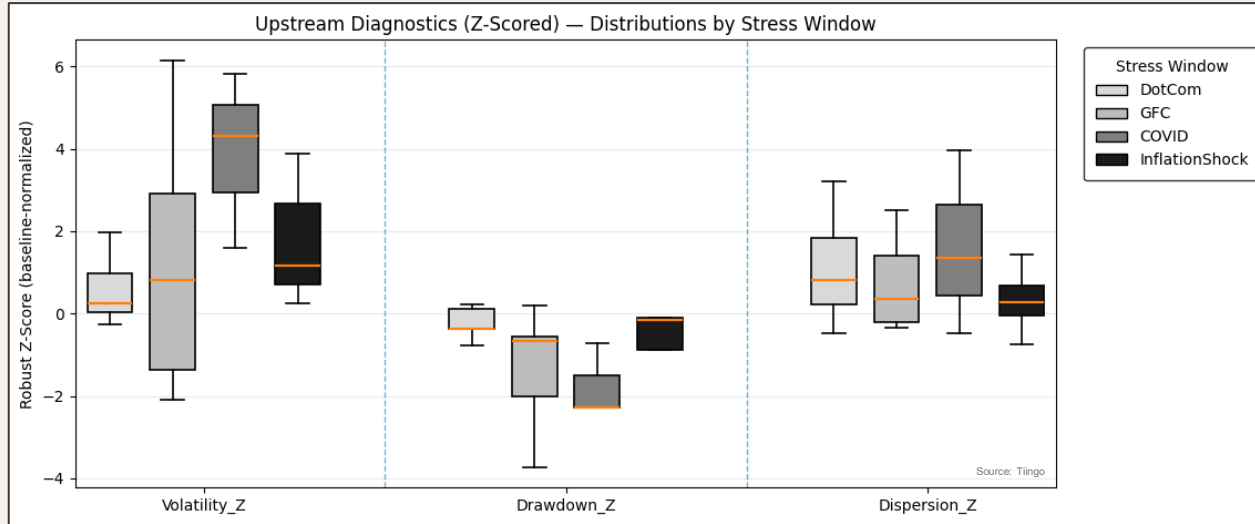
Cert. Data Science | Springboard, **2021**

B.A. Economics | Michigan State University, **2009**



Appendix: Z-Score Distributions by Regime

Robust Diagnostics Under Stress (Baseline-Normalized)



Diagnostics as Robust Z-scores (median / IQR), normalized to long-run baseline;
Volatility for rolling dispersion, **Drawdown** for rolling peak-to-trough loss, &
Dispersion for cross-sectional spread.

Distributional diagnostics confirm regime-level medians shown in the main heatmap are representative, while preserving visibility into dispersion, skew, & tail behavior in stress regimes.